

Video Transcript

Video 6: Packages in Java (04:40)

One of the issues you're going to run into is that of complexity as your code grows. Now a very effective technique to be able to address this is to be able to break things apart to minimize the complexity, the amount of code that you have in any one file. Now we did this previously, however, we leveraged the implicit resolution that exists within the framework, within the language to be able to do that and we kept all the files in one directory. Now that doesn't scale very well. And what you want to do is you want to leverage some of what has been built in into the language. And as you can see here, you can do packages and then those packages can be imported as I'm illustrating here, within these two files.

You can see here that the person file at the top is defining a package called university. Now on the right-hand side you can see there that file is then importing that package in that class as you see there. Now what this means from a practical standpoint when it comes to the file system is that the university package is going to live inside of a directory called University. As you can see there above that panel that defines the code and that the consumer class is on a directory above. So, I'm explicitly saying this within that node at the right-hand side at the bottom that says person must be in subdirectory named University.

So, let's go ahead and implement this as I think it will be clearer than simply describing it. So, as you can see here, I have those files already within the directory. I'm going to go ahead and take a look at Person. Person has the code we expect it to have. If I look at Consumer, the same thing is true. You can see there that as well. Now we need to add that package definition and we need to create those directories. So, I'm going to go ahead and start here by creating that directory. So, I'm going to call a directory. I'm going to create a directory called University.

Then I'm going to go ahead and move the Person file into it, into University. Now if we look at it, it has been moved into it, and now we're going to go ahead and move into University. We're going to go ahead and confirm that, that file is there, and now I'm going to go ahead and open up that directory. Now the thing that we want to do here is we want to define that package, and the package is going to be called University, which is the name of the directory that it's in. I'm going

to go ahead and then save that file, then confirm that, that change has been added. And you can see there that it has. I'm now going to go ahead and compile it. So, I'm going to go ahead and enter Javac, then Person. And as you can see there that has been compiled. I'm now going to move a directory back up. And so I'm back at consumer and you can see there the directory called University. You can also list it this way so that it looks a little bit easier to read.

And now we're going to have to add those changes to Consumer. So, let's go ahead and open up that file. And here we're going to import. We're going to use the directory name University.Person, and then the rest of it is the same. All we're doing is adding visibility and bringing in that package. So, I'm going to go ahead and close, save, I'm going to go ahead and compile, in this case, Consumer. As you can see there, we have it already. So, let's go ahead and run it to confirm that what we did actually worked. So, let's go ahead. And as you can see there, we get the same result as before. That class is instantiated. That means we can import that package and then we can execute functionality within that new instance of the class. So, now it's your turn. Make sure you understand how packaging works within the Java framework and that you can repeat what I have done.

